DocHMET: Synergizing Document Intelligence and Multi-Agent Work-flows for Automated HMET Model Generation

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***Abstract*—Accurate device modeling is essential for optimizing the performance of HEMTs, but traditional manual parameter extraction from documents is time-consuming, labor-intensive, and prone to errors. This study introduces DocHEMT, a novel automated system that leverages advanced artificial intelligence (AI) technology to extract key electrical parameters from circuit documents and generate corresponding SPICE models. The system comprises three main components: document parsing and curve decomposition, dataset Retrieval-Augmented Generation (RAG) and parameter extraction, and SPICE modeling. By integrating data extraction models, large language models (LLMs), and optimization techniques, DocHEMT significantly improves the efficiency and accuracy of device modeling, reducing the need for manual intervention. The effectiveness of DocHEMT is demonstrated through the processing of the FHX76LP datasheet, showcasing its ability to convert complex graphical information into structured curve data and generate accurate SPICE models.**

***Index Terms*—Retrieval-augmented Generation, ASM-HEMT**

I. Introduction

High Electron Mobility Transistors (HEMTs) hold very broad applications [1]. HEMT device modeling is crucial for electronic design and performance optimization. With the continuous development of technology, extracting key parameters from various documents and charts to build accurate device models has become a challenging task. Traditional methods often rely on manual extraction, which is not only time-consuming and laborious but also prone to errors.

In recent years, large models have achieved significant results in the fields of natural language processing and image recognition [2-3]. These large models have strong feature extraction and learning capabilities, providing new possibilities for the automatic extraction of data. By leveraging the advantages of large models, efficient and accurate parameter extraction can be achieved, thereby improving the quality and efficiency of device modeling.

However, there are still some issues in applying large models to chart parsing and device modeling, such as how to ensure that the model can accurately identify and extract key information from charts, and how to deal with different types and formats of charts. Therefore, research on device modeling based on large-model chart parsing has important practical significance, and is expected to bring new development to the field of electronic design automation.

To address these challenges, we have introduced DocHEMT, a novel automated system that utilizes advanced artificial intelligence technology to extract key electrical parameters from circuit documents and generate corresponding SPICE models. The system includes the following key components:

(a) Document Parsing: The first step of the DocHEMT system is document parsing and curve decomposition. This involves analyzing circuit documents to identify and extract relevant information. The system uses natural language processing techniques to understand the text content of the documents and identify key parameters related to HEMT devices. In addition, the system also uses image recognition algorithms to extract characteristic curves from charts. These curves are then decomposed into individual data points for further analysis and modeling.

(b) Dataset RAG: After the document parsing, the system generates a list of ASM-HEMT parameters and their initial value ranges. This is accomplished through a process known as Knowledge Base RAG (Retrieval-Augmented Generation). The system constructs a domain knowledge base using a large dataset of HEMT-related documents and charts. The model then retrieves and generates corresponding context based on the information extracted from the input document, which is combined and input into the LLM to generate a list of parameters and initial value ranges, providing a starting point for the SPICE model generation process.

(c) SPICE Extraction: DocHEMT initiates the model extraction process using the extracted parameters and initial value ranges, optimizes the ASM-HEMT parameters, and generates SPICE models that best match the datasheet specifications.

DocHEMT integrates the powerful capabilities of data extraction models, LLMs, and optimization techniques, providing a viable path for the automation of device modeling. It not only simplifies labor-intensive manual operations but also significantly improves the efficiency and accuracy of converting document data into device models. In the following research and experiments, we will provide a detailed introduction to the principles and implementation of each module of DocHEMT.

II. Parameter Extraction using LLM

The architecture of the proposed DocHEMT is shown in Fig.1. Firstly, it employs a specialized device document parsing model to effectively segment different components of the document, including graphics, curves, parameters, and circuit diagrams. Then, a curve recognition model is used to extract graphic curves and convert them into values. Subsequently, a domain knowledge base is constructed using a large dataset of HEMT-related documents and chart data. The model retrieves and generates corresponding context based on the information extracted from the input document, which is then combined and input into the LLM to generate a list of ASM-HEMT parameters and initial value ranges. Finally, the spice model is optimized. Through this automated process, DocHEMT can efficiently and accurately convert circuit datasheets into SPICE models, greatly reducing the need for manual intervention.

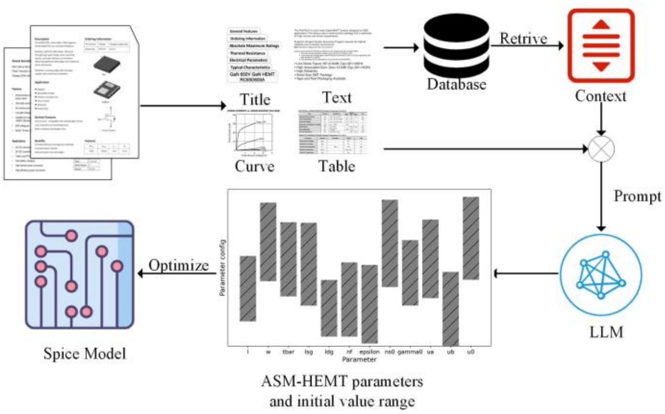


Fig. 1 The whole framework of the DocHEMT

A. Document Parsing and Curve Decomposition

Given the diverse structures and high information dimensions of electronic device documents, we have adopted a multi-layer depth-wise separable convolutional network architecture, integrated with Squeeze-and-Excitation (SE) modules, to enhance the interaction and learning of features between channels. Firstly, EDocNet employs depth-wise separable convolution, which significantly reduces computational complexity while maintaining its expressive power. Additionally, by incorporating lightweight activation functions such as H-swish, the model further reduces the computational overhead of inference time, enabling faster inference speeds and higher accuracy even on resource-constrained hardware. During model training, EDocNet adopts a fusion of local focus and global knowledge distillation strategies (FGD). This approach guides the model to capture fine-grained details (such as device models and parameter values) as well as the overall document structure, effectively segmenting different components of the document, including graphics, curves, parameters, and circuit diagrams.

For the extracted feature curve images, the YOLOv8 model is used to detect and locate curve charts, axes, legends, labels, and other graphical elements within the document. After localization, OCR is applied to recognize text annotations within these areas, such as axis labels, scale lines, and legend descriptions, which are crucial for understanding the context, units, and coordinate systems of the curves. Subsequently, we employ an instance segmentation model based on the encoder-decoder Transformer architecture, enhanced by a masked attention pixel decoder. After extracting the curves, we map the image coordinates to actual physical coordinates. By aligning the extracted curves with the recognized axis scales, we generate high-precision data points that can be directly used for simulation or model fitting. Through this complete automated process, our framework can efficiently convert complex graphical information in data sheets into structured curve data.

B. Dataset RAG

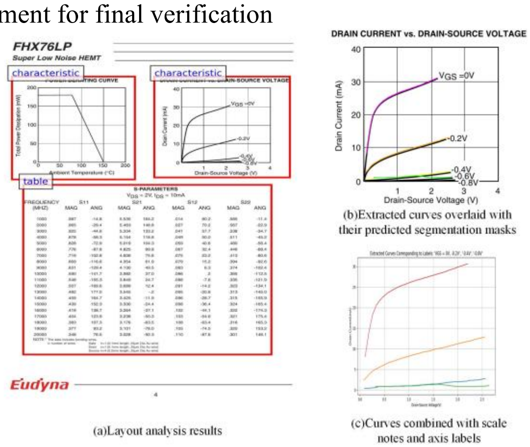
The system constructs a domain knowledge base based on a large number of HEMT-related documents, papers, and datasheets. First, various forms of documents are segmented into different chunks, which are then embedded into a vector - form knowledge database. After completing the parsing of technical documents and the extraction of device characteristic curves, the system retrieves relevant contextual information from the vector database. This information, along with the parsing results and curve extraction results, is fed into the large model. The LLM will invoke prior knowledge from the field of circuit design and integrate conclusions formed in earlier reasoning steps to verify and correct ambiguous information, thereby ensuring the accuracy and reliability of the extracted information. A subset of parameters that need to be optimized is carefully selected from the ASM - HEMET model library, which contains more than 200 adjustable parameters. By precisely matching the extracted device characteristics (such as threshold voltage, mobility factor, etc.) with the corresponding modeling coefficients, the system can ensure that only parameters significantly affecting device performance are extracted.

C. SPICE Extraction

The ASM-HEMT [3] is an industry-standard SPICE model for HEMTs, standardized by the Compact Model Coalition (CMC) in 2018. Since then, it has become the mainstream model. In this study, a multi-stage optimization method based on the Optuna framework is adopted to extract SPICE model parameters. First, using the initial parameter range of ASM-HEMT generated by the knowledge base RAG from the previous stage, combined with Optuna's dynamic Bayesian optimization algorithm, the parameter space is efficiently explored to find the optimal model parameters that best match the experimental data. During the optimization process, we use distributed measurement to improve search efficiency and optimization accuracy. In the first stage, the parameter range given in the previous step is expanded to ensure coverage of the potential optimal solution space. On this basis, the second stage further narrows the parameter search range for more refined optimization. This strategy not only improves optimization efficiency but also ensures the accuracy and reliability of the extracted parameters.

III. Experiment Results

To verify the effectiveness of DocHMET, the datasheet of FHX76LP was processed as an example. This device is a low - noise HEMT designed for high - frequency and high - gain applications, and its datasheet contains a large amount of text descriptions, tables, and multiple performance curves, which have high reference value. DocHMET was executed on a workstation equipped with two Nvidia GTX 4090 GPUs. A HEMT - domain knowledge base was constructed based on Tencent Yuanqi, and the deepseek - R1 large model was invoked. SPICE simulation was carried out in the NGSPICE environment for final verification



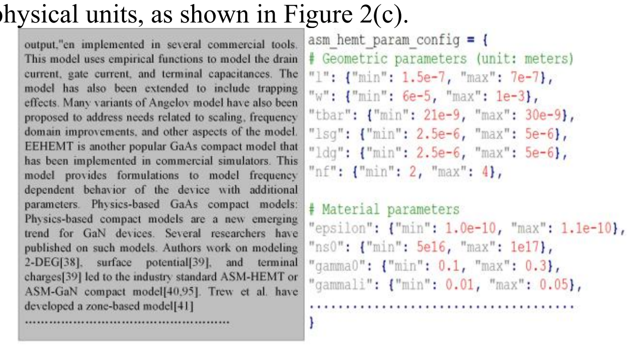
(a)Layout analysis results

(b)Extracted curves overlaid with their predicted segmentation masks

(c)Curves combined with scale notes and axis labels

Fig. 2 Document parsing and curve decomposition results

First, the FHX76LP datasheet was subjected to document segmentation and curve extraction, as shown in Figure 2(a), where various parts of the document can be effectively segmented. Subsequently, YOLOv8 was used to perform object detection and localization of elements such as axes, legends, and labels in the graphics, providing reference points for subsequent coordinate mapping. To further refine the extracted data, trajectory tracking technology was used to convert the segmented curves into discrete coordinate points. Then, axis matching and unit recognition were carried out to convert the pixel - based coordinates into physical units, ensuring consistency with the scale markings in the datasheet, as shown in Figure 2(b). Finally, by integrating object detection, OCR, and curve extraction, the workflow generated DC I - V curves with physical units, as shown in Figure 2(c).



(a)Knowledge base RAG result

(b)Parameter list generation

Fig. 3 RAG and parameter list generation

This study develops a knowledge base architecture based on a large language model. The knowledge base integrates over 100 documents, including research papers on HEMTs from mainstream academic databases, official ASM-HEMT documentation, and technical documents from major manufacturers, with a total size of approximately 300 MB. When specific device experimental data is input, the system retrieves relevant entries from the knowledge base using a hybrid retrieval mechanism and generates contextual information, as shown in Figure 3(a). The LLM then refines the retrieved results, guided by a prompt template designed to extract core parameters and their recommended ranges for the ASM-HEMT model based on typical parameter combinations found in the literature, as illustrated in Figure 3(b).

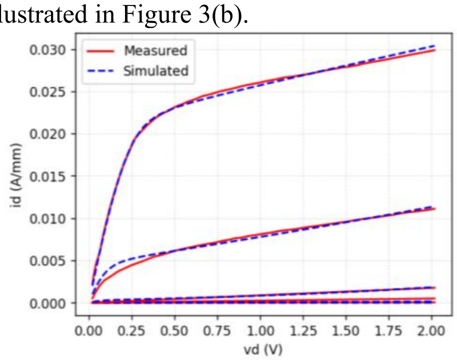


Fig. 4 Model Extraction Optimization results

The optimization of the SPICE model is performed based on the initial parameter ranges extracted using the RAG and LLM methods, as shown in Figure 4. The optimized ASM-HEMT model achieves a Normalized Root Mean Square Error (NRMSE) of 0.934% compared to the actual parameters. The results demonstrate that the optimized model parameters are accurate and reliable.

IV. Conclusion

This study presents a comprehensive solution to the challenges of device modeling for HEMTs by introducing DocHEMT, an automated system that utilizes AI and LLMs to extract parameters from circuit documents and generate SPICE models. The system’s key components, including document parsing and curve decomposition, dataset RAG, and SPICE modeling, work together to enhance the efficiency and accuracy of the modeling process. The successful application of DocHEMT to the FHX76LP datasheet demonstrates its potential to significantly reduce manual effort and improve the quality of device modeling.

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